

Investigating the Impact from Ultrasonic Testing Uncertainty in Quantifying Ply Stack Orientation on the Probabilistic Failure Envelope



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Introduction

Ultrasonic testing (UT) is of interest in the present study due to its affordability where others need a higher cost of investment. CFRP Laminate Properties depend mostly on lamina orientation. This study investigates sensitivity of the expected performance on the confidence of the ply stack, based upon the work of Blackman and Jack who used 2D FFT technique to predict each ply orientation with an average accuracy within 2 degrees for 98% of the lamina[1]. Implementing uncertainty in the measured ply stack orientation, and Tsai-Wu failure envelope the probabilistic Cumulative-Density-Function (CDF) of First Ply Failure (FPF) stresses using Monte-Carlo Simulation technique are presented.



Modeling of the Composite

- Lamina and Laminate coordinate system is indicated by $(\bar{x}_1, \bar{x}_2, \bar{x}_3)$ and (x_1, x_2, x_3) of k^{th} lamina.
- ϕ_k is the rotation angle between the principal axis.

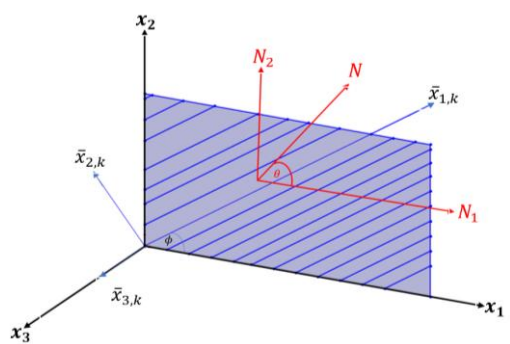
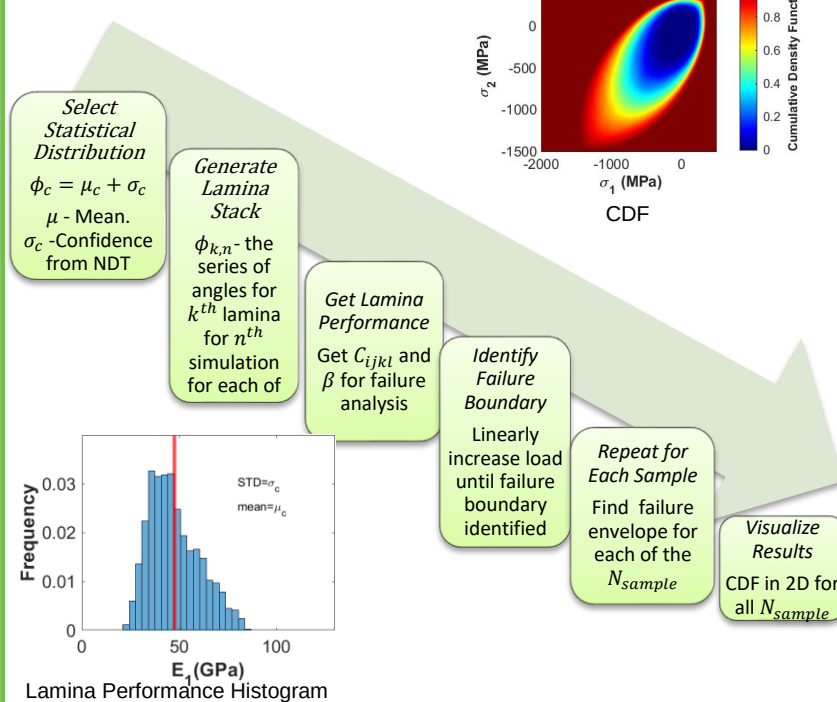


Figure 1 k^{th} lamina of the composite in principal and global coordinate

- For unidirectional lamina properties, reference from Narayanan [2] is used.

Monte Carlo Simulation



Ultrasonic-Testing (UT) is one of the more popular techniques among Non-Destructive-Testing (NDT) for the inspection of manufactured and in-service components in automotive and aerospace industries. With the advancement of NDT waveform analysis to determine the ply-orientation of Carbon-Fiber-Reinforced-Polymers (CFRP) by recent researchers, the ability to determine the individual ply-orientation on a lamina-by-lamina basis has improved from an accuracy of $\pm 10^\circ$ to $\pm 2^\circ$. This work investigates the impact of this improvement in resolution quantification on the confidence in characterizing the stiffness and strength of CFRP laminates. This is done by stochastically generating ply stack orientation from UT inspection results and quantifying the failure envelope from the Tsai-Wu failure model through a probabilistic Cumulative-Density-Function (CDF) of First Ply Failure (FPF) stresses using a Monte-Carlo Simulation technique. A variety of planar loads are generated to form the failure envelope for CFRP laminates consisting of 3, 8, 12 and 16 plies for a variety of layup sequences. As the ply stack quantification from UT data is stochastic, so is the confidence of the failure envelope. A comparative analysis of the stochastic failure envelope using the previous UT confidence ($\pm 10^\circ$ for each individual lamina) and the recent UT analysis method confidence ($\pm 2^\circ$ for each individual lamina) is provided to show the improvement in characterizing the probabilistic failure envelope, and results show a significant and measurable improvement in the confidence in the as manufactured performance of the composite laminate.

Materials and Manufacturing

Standard Stacking sequence and configurations from Blackman and Jack [2] is considered for analysis. Parts 8C, 16C and 12M are a unidirectional composite, whereas Part 3B is a woven laminate.

Table-1: Laminate Stacking Sequence

Part Name	Number of Plies	Orientation
8C	8	[0/90/0/90/0/90/0/90]
16C	16	[0/90/0/90/0/90/0/90/0/90/0/90/0/90/0/90]
12M	12	[0/30/-30/45/-45/90/90/-45/45/-30/30/0]
3B	3	[10/18/27]

Results and Discussion

Effective Stiffness

- $N_{sample} = 1,000$ is used to generate 1,000 randomly sampled laminate layups. The standard deviation λ_k is set to 2° based upon the high-resolution inspection data found in [1], whereas the value for 10° is based upon the earlier studies by the group identified in [1].
- The histogram of the effective moduli for Sample 12M is shown in Figure 2 for E_1 .

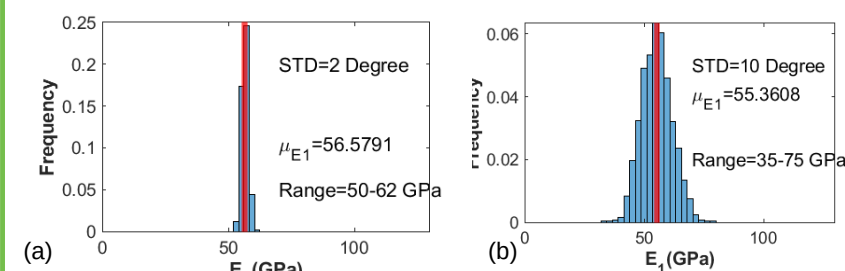


Figure 2 Variation of Effective Engineering Properties, E_1 (Sample 12M) (a) 2° deviation, (b) 10° deviation

Failure Envelope

- The Cumulative Density Function (CDF) plot is generated for the longitudinal σ_1 and transverse σ_2 stresses presents the stochastic variation of the failure probability. Figure 3, indicates the safe region for loading by the blue color and the loading states that would cause a failure in red.

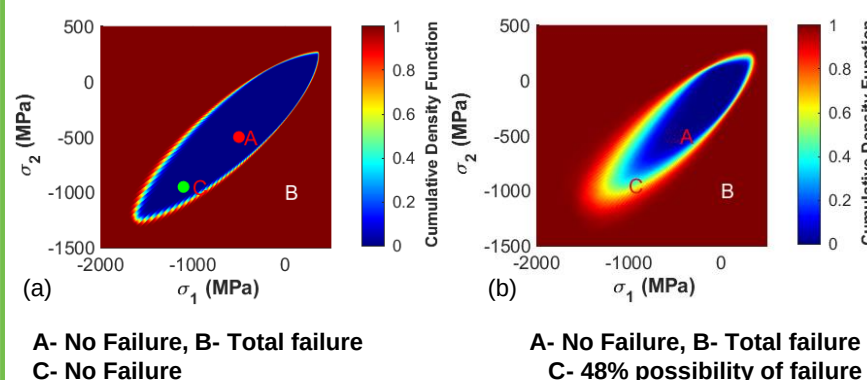


Figure 3 CDF variation of failure envelope (Sample 12M) (a) 2° confidence in inspection, (b) 10° confidence in inspection

Failure Envelope

- The sensitivity to the ply variability decreases with increasing number of lamina in the stack.

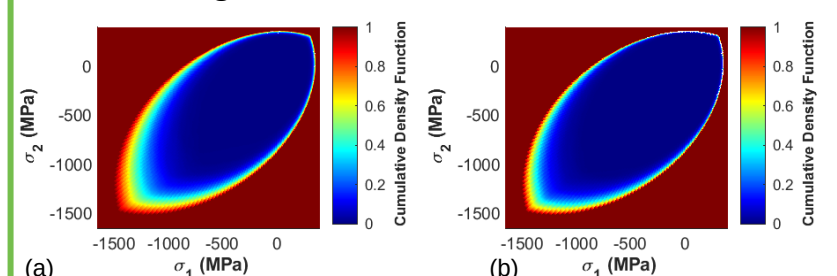
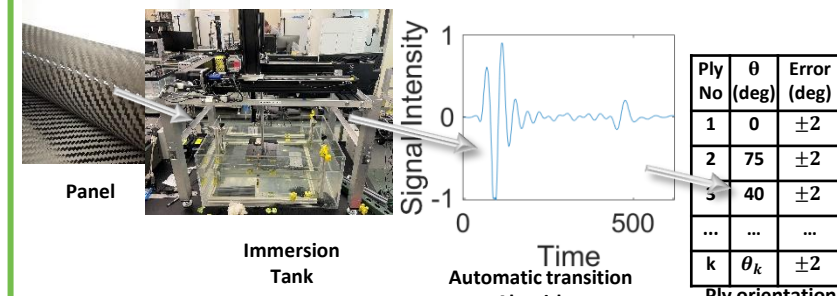


Figure 4 Narrower Failure Envelope with increasing ply numbers, with a 2° confidence in ply stack orientation (a) Sample 8C, (b) Sample 16C

NDT Part Analysis

The ply orientation data is generated from high resolution pulse echo UT and later implementing FFT for CFRP with plain weave in [1].



CDF of Sample 3B from [2] shows different failure envelope shape than traditional ply stacks studied above. Regardless, the conclusion is the same.

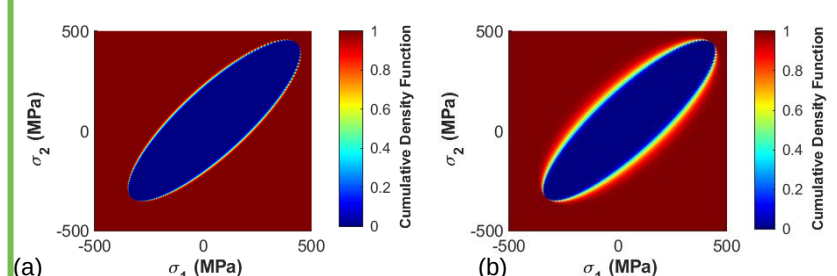


Figure 5 CDF variation of failure envelope (Sample 3B) for a (a) 2° and (b) 10° confidence in inspection

Conclusion

Part	σ_1	Max σ_1 (Mpa)		σ_2	Max σ_2 (Mpa)	
		2°	10°		2°	10°
8C	1150	1505	1567	1150	1506	1521
16C	1312	1505	1512	1312	1505	1508
12M	1672	1722	1835	1260	1304	1497
3B	451	457	457	451	489	489

Acknowledgements

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Reference

- Blackman and Jack, Under review, "Novel Technique for the Automated Determination of the Ply Stack for a Woven Carbon Fiber Composite using High-Frequency Pulse Echo Ultrasound," Sep. 08, 2021.
- Narayanan, "Microwave Nondestructive Testing of Galvanic Corrosion and Impact Damage in Carbon Fiber Reinforced Polymer Composites," *International Journal of Microwaves Applications*, 7(1):1-15, 2018.

Results from an inspection with a confidence of 2° would lead the analyst to conclude that a loading condition was safe, whereas a confidence of 10° would may conclude the same loading condition was not safe. Thus, having access to lower quality inspection data will lead one to either repair or scrap a functional and safe part unnecessarily.